iRobot Web Controller Setup Guide

Prerequisites

- Raspberry Pi with ROS2 installed
- iRobot Create 3 connected to Raspberry Pi
- Python 3.8 or higher

Step 1: Install Python Dependencies

Create a `requirements.txt` file:

```txt
flask==2.3.0
flask-cors==4.0.0
```

Install dependencies:

```bash
pip3 install -r requirements.txt
```

Step 2: Create Project Structure

```bash
mkdir ~/irobot_web_controller
cd ~/irobot_web_controller
mkdir static
```

Your folder structure should look like:

```
~/irobot_web_controller/
├── server.py          # Backend server (from artifact)
├── requirements.txt   # Python dependencies
└── static/
    └── index.html     # Web interface
```

Step 3: Save the Files

1. **Save the backend server:**
   - Copy the `server.py` code from the artifact above

- Save it as `~/irobot_web_controller/server.py`
- Make it executable: `chmod +x server.py`

2. **Save the web interface:**
   - Go back to the web controller artifact
   - Copy the entire React component
   - Save it as `~/irobot_web_controller/static/index.html`

Step 4: Configure ROS2

Before running the server, source your ROS2 workspace:

```bash
source /opt/ros/<your-ros-distro>/setup.bash
source ~/create3_examples_ws/install/local_setup.sh

# Set safety override (as in your script)
ros2 param set /motion_control safety_override full
```

Step 5: Update the Web Interface to Use Backend

In your `index.html`, uncomment the API calls and update the server URL:

Replace this line in the React code:

```javascript
// TODO: Replace with actual API call to Raspberry Pi
```

With:

```javascript
fetch('http://<RASPBERRY_PI_IP>:5000/api/cmd_vel', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ linear, angular })
});
```

Step 6: Run the Server

```bash
```

```
cd ~/irobot_web_controller
python3 server.py
```

You should see:

```
============================================================
iRobot Web Controller Server
============================================================
Server starting on http://0.0.0.0:5000
Make sure ROS2 is sourced and running!
============================================================
```

Step 7: Access the Web Interface

1. Find your Raspberry Pi's IP address:

```bash
hostname -I
```

2. Open a web browser on any device on the same network
3. Navigate to: `http://<RASPBERRY_PI_IP>:5000`

Testing

1. Click "Connect" button
2. Test the joystick - robot should move
3. Try "Dock" and "Undock" buttons
4. Test emergency stop

Troubleshooting

Server won't start

- Check if port 5000 is already in use: `sudo lsof -i :5000`
- Try a different port in `server.py`: `app.run(host='0.0.0.0', port=8080)`

Robot won't move

- Check ROS2 is running: `ros2 node list`
- Check topics: `ros2 topic list`
- Verify safety override: `ros2 param get /motion_control safety_override`

- Monitor velocity commands: `ros2 topic echo /cmd_vel`

Can't connect from browser

- Check firewall: `sudo ufw allow 5000`
- Verify Raspberry Pi IP address
- Make sure devices are on same network

Commands timeout

- Increase timeout in `server.py`
- Check ROS2 daemon: `ros2 daemon status`

Running on Startup (Optional)

Create a systemd service:

```bash
sudo nano /etc/systemd/system/irobot-web.service
```

Add:

```ini
[Unit]
Description=iRobot Web Controller
After=network.target

[Service]
Type=simple
User=pi
WorkingDirectory=/home/pi/irobot_web_controller
ExecStart=/usr/bin/python3 /home/pi/irobot_web_controller/server.py
Restart=always

[Install]
WantedBy=multi-user.target
```

Enable and start:

```bash
sudo systemctl enable irobot-web.service
sudo systemctl start irobot-web.service
sudo systemctl status irobot-web.service
```

Security Notes

⚠️ **Important:** This setup is for local network use only!

For production:

- Add authentication
- Use HTTPS
- Implement rate limiting
- Add input validation

Next Steps

Once basic control works, you can add:

- STM32 sensor data display
- Battery status
- Camera feed
- Advanced navigation features
- Multi-robot control